UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 5 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. A vector image becomes pixelated when enlarged.
2. Frame rate is the number of video frames displayed per second.
3. Interactivity level is irrelevant when choosing an authoring tool.
4. CMYK is used for printed output.
5. A storyboard is a visual plan of a multimedia sequence.
6. A linear presentation allows users to choose their own path.
7. A multimedia project always uses every media element.
8. Interactivity gives users control over content and flow.
9. A higher sampling rate generally produces a more accurate recording.
10. Interlacing displays every line of a video frame at the same time.
11. Hypermedia combines multimedia with hyperlinks.
12. Sampling converts a continuous sound wave into digital values.
13. Multimedia combines text, graphics, audio, video and animation.
14. USB is a plug-and-play connection standard.
15. Kerning adjusts the space between individual character pairs.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A. Matching options

A. Serif	B. Sampling	C. Cel animation
D. Hypermedia	E. Tweening	F. Bitmap
G. Linear multimedia	H. Browser	I. Keyframe
J. Scanner	K. Target audience	L. Plotter
M. Storyboard	N. Multimedia	O. Frame rate

1. font with small strokes at character ends
2. visual sequence plan
3. input device that digitises documents
4. important frame in an animation
5. frames displayed per second
6. output device for large drawings
7. image made of pixels
8. generating in-between frames
9. intended users of a product
10. software that displays web pages
11. converting sound into digital values
12. integration of text, graphics, audio, video and animation
13. presentation without user control of sequence
14. linked multimedia elements
15. hand-drawn frame-by-frame animation

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PART C (30 MARKS)

Choose the correct answer for each question.

1. Which is NOT a multimedia element?
   A. Animation
   B. Compiler
   C. Text
   D. Audio

2. A hyperlink links to:
   A. Only images
   B. Only text
   C. Another resource
   D. Only audio

3. A higher sampling rate generally:
   A. Improves accuracy
   B. Reduces accuracy
   C. Changes colour
   D. Removes audio

4. Amplitude of a sound wave relates to:
   A. Pitch
   B. Loudness
   C. Frame rate
   D. Colour

5. Tweening creates:
   A. Sound effects
   B. In-between frames
   C. Text fonts
   D. Video codecs

6. MPEG is:
   A. A storage device
   B. A font
   C. A video compression family
   D. An image editor

7. Plug and play means:
   A. Automatic device recognition
   B. No power needed
   C. Wireless only
   D. Manual driver installation only

8. Which image type scales without losing quality?
   A. Vector
   B. Scan
   C. Photograph
   D. Bitmap

9. Hypertext contains:
   A. Links to other content
   B. Only images
   C. No text
   D. Only audio

10. A browser is used to:
    A. Edit audio
    B. Print documents only
    C. Display web pages
    D. Scan images

11. MIDI stores:
    A. Digitized audio samples
    B. Video frames
    C. Musical notes and instructions
    D. Images

12. Cel animation is created by:
    A. Printing images
    B. Recording sound
    C. Drawing each frame by hand
    D. Scanning text

13. Copyright protects:
    A. File names
    B. Original creative works
    C. Internet speed
    D. Computer hardware

14. Chroma key is used to:
A. Compress audio
B. Replace a colour background
C. Add text
D. Scan images

15. Streaming allows media to:
A. Only play offline
B. Require full download first
C. Play while downloading
D. Remove compression

16. Anti-aliasing is used to:
A. Smooth jagged edges
B. Increase file size only
C. Add sound
D. Compress video

17. A storyboard is:
A. A visual sequence plan
B. A video codec
C. A sound file
D. A colour model

18. Path animation moves an object:
A. Randomly
B. Along a predefined route
C. Only vertically
D. Without motion

19. Which is an authoring tool category?
A. CMYK
B. JPEG
C. Card-based
D. RGB

20. Virtual reality creates:
A. A printed page
B. A simulated environment
C. A text font
D. A camera lens

21. HTML is used to:
A. Structure web content
B. Model 3D objects
C. Compress video
D. Store audio samples

22. A graphics tablet is used for:
A. Sound output
B. Drawing input
C. Printing
D. Web browsing

23. An authoring tool is used to:
A. Only print text
B. Only record audio
C. Assemble multimedia elements
D. Only browse the web

24. Which format is commonly used for bitmap images?
A. MPEG
B. PNG
C. HTML
D. MIDI

25. A bitmap image is made of:
A. Mathematical objects
B. Pixels
C. Sound samples
D. Text characters

26. Which is a 3D animation task?
A. Sampling
B. Bookmarking
C. Modelling and rendering
D. Kerning

27. Card-based authoring tools are suited to:
A. 3D modelling only
B. Messages with a beginning and end
C. Audio editing only
D. Live video only

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28. Which is an input device?

A. Plotter
B. Printer
C. Scanner
D. Monitor

29. Which best defines multimedia?

A. A printed book
B. Text, graphics, audio, video and animation
C. Text only
D. A single image

30. Resolution refers to:

A. Colour name
B. Number of pixels in an image
C. Font size
D. File type

PART D (40 MARKS)

QUESTION 1

You are required to animate a company logo. The logo depicts a planet orbiting the sun.

a) State TWO animation file formats suitable for this project.

(2 marks)

b) Describe the steps of cel animation and ONE advantage of using it.

(4 marks)

c) Differentiate 2-D and 2.5-D animation with ONE example for each.

(4 marks)

QUESTION 2

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

(3 marks)

b) Describe TWO roles of hyperlinks in the website.

(3 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

(4 marks)

QUESTION 3

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(3 marks)

b) Explain the interactivity level suitable for the demonstration.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(4 marks)

QUESTION 4

A product catalogue must show sharp images on screen and in print.

a) State TWO differences between bitmap and vector images.

(2 marks)

b) Describe TWO factors that affect image quality.

(3 marks)

c) Explain how colour model and resolution affect screen and print output.

(5 marks)

END OF QUESTION PAPER